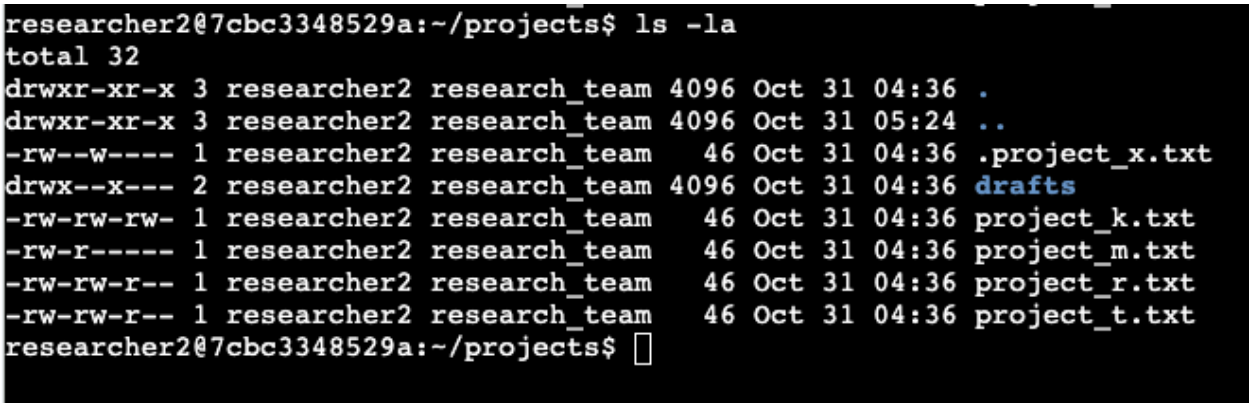
permissions in Linux

Project description

This project will review who can access the research team's files and check that their permissions match their roles. Where access is outdated or hasn't been approved, permissions will be updated or removed. The goal is to protect the organization's information while making sure staff can access the files they need.

Check file and directory details

I used the Linux command `ls -la` to review the files and folders in the `~/projects` directory. The output showed their permissions, owners, and groups, which helped me check who could read, write, or access each item and identify permissions that might need to be updated.



The first line shows the command I entered, `ls -la`, and the lines below show its output. I ran the command from the project directory to list its contents, including hidden items, with details such as permissions and ownership. The listing shows one hidden file, `.project_x.txt`, a directory named `drafts`, and four other files: `project_k.txt`, `project_m.txt`, `project_r.txt`, and `project_t.txt`. The 10-character field at the start of each entry shows the item type followed by the read, write, and execute permissions for the owner, group, and others.

Describe the permissions string

For `project_r.txt`, the permissions string is `-rw-rw-r--`. The first character, `-`, indicates that this is a regular file. The next three characters, `rw-`, show the owner can read and write the

file but cannot execute it. The following `rw-` shows the same permissions for the group. The last three characters, `r--`, mean other users can read the file but cannot write or execute it. In the string, `r` means read, `w` means write, `x` means execute, and `-` means that permission is not granted.

Change file permissions

```
researcher2@7cbc3348529a:~/projects$ chmod o-w project_k.txt
researcher2@7cbc3348529a:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 04:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 05:24 ..
-rw--w---- 1 researcher2 research_team   46 Oct 31 04:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 31 04:36 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 31 04:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_t.txt
researcher2@7cbc3348529a:~/projects$
```

I used `chmod o-w project_k.txt` to remove write permission for others `(o)` from `project_k.txt`. The command completed without displaying an error. I then ran `ls -la` to check the permissions. The listing showed `project_k.txt` changed from `-rw-rw-rw-` to `-rw-rw-r--`: the owner and group still have read and write access, while others can only read the file.

Change file permissions on a hidden file

```
researcher2@7cbc3348529a:~/projects$ chmod u-w,g+r,g-w .project_x.txt
researcher2@7cbc3348529a:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 04:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 05:24 ..
-r--r----- 1 researcher2 research_team   46 Oct 31 04:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 31 04:36 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 31 04:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_t.txt
researcher2@7cbc3348529a:~/projects$
```

Because `.project_x.txt` is an archived, hidden file, the research team wants to prevent

anyone from changing it while keeping it readable by the owner and group. I ran `chmod u-w,g+r,g-w .project_x.txt` to remove write permissions and give the group read access, then used `ls -la` to verify the change. The listing showed `-r--r-----`, confirming that the owner and group can read the file, no one can write to it, and others have no access.

Change directory permissions

```
researcher2@7cbc3348529a:~/projects$ chmod g-x drafts
researcher2@7cbc3348529a:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 04:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 31 05:24 ..
-r--r----- 1 researcher2 research_team   46 Oct 31 04:36 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Oct 31 04:36 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 31 04:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 31 04:36 project_t.txt
researcher2@7cbc3348529a:~/projects$
```

The `drafts` directory belongs to `researcher2`, and the research team should not be able to access it. I used `chmod g-x drafts` to remove the group's execute permission, which prevents group members from opening or searching the directory. I then ran `ls -la` and confirmed the permissions were `drwx------`, meaning `researcher2` has access while the group and others do not.

Summary

I used `ls -la` to review the files and folders in the projects directory, including their ownership and permissions. I removed write access for others on `project_k.txt`, set `.project_x.txt` so the owner and group can read it but no one can write to it, and restricted the `drafts` directory to `researcher2`. These changes support the organization's goal of giving the research team only the access needed for their work.